06C — Deep-Dive Feature Audit: Pills & Alchemy (Part 3 of 6)

Realm Breakthrough Pills (`P:BT_*`), Major Bottleneck Treasures (`BT.TREASURES`), Drug Resistance (`btTaken`), Wrong-Path Toxicity (`×1.8`) & Triple `btBonus` Variable Split

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Breakthrough preparation, pill synthesis, drug-tolerance calculation, major-realm treasure hunting, and vow adherence must arise naturally across **all 43 Major Paths** without orphaned demand queues, bypassed resistance rules, or split bonus fields.
- **Immutable Project Rules**:
    1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
    2. **Preserve exact 40 `<script>` tags** and load order in `index.html` / `test/load.js`.
    3. **Strict Closed-Loop Resource Conservation** & **Zero RNG consumption in read/score/UI paths**.

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1. Architectural Scope of Group 3 (`06C`)

Group 3 covers every pill, treasure, and preparation pipeline governing **Major-Realm Bottlenecks (คอขวดทะลวงด่านใหญ่)**:

1. **Path- & Stage-Specific Breakthrough Pills (`P:BT_<PATH>_<STAGE>`)**:
    - Original 3 Pillars (`data_bt.js`): `BT_XIAN_1..8` (8 pills), `BT_BODY_1..4` (4 pills), `BT_GOD_1..3` (3 pills).
    - First-Wave Expansions (`demon_heartsoul_paths.js` / `data_cult.js` / `data_expansion.js`): `BT_DEMON_*`, `BT_HEART_SOUL_*`, and partial expansion entries.
    - **31 Newer Grand Dao Paths (`v2.0.0` + `v2.2.0`): `0` dedicated `BT_<PATH>_<STAGE>` recipes (`0 / 217` stage transitions!)**.
2. **General Non-Path-Locked Breakthrough Pills**:
    - `REC140_ALC_FOUNDATION_3` (โอสถสร้างรากฐานจิตหยก, `+18%`), `REC140_ALC_GOLDEN_CORE_4` (โอสถควบจินตันเพชรทอง, `+22%`),

ECO2_REC_PILL_CULT_QI_*, REC140_PARTIAL_DRAGON_PILL_7 (+35%), and ITEM_V234_* (EFX_BREAKTHROUGH_BOOST).

3. **Major-Realm Bottleneck Treasures (JD.BT.TREASURES in data_bt.js & bottleneck.js:13)**:
   - TR_FOUNDATION_JADE (XIAN:3), TR_CORE_CRYSTAL (XIAN:5), TR_SOUL_LOTUS (XIAN:7), TR_BODY_BLOOD (BODY:2), TR_DRAGON_MARROW (BODY:4), TR_FAITH_SEAL (GOD:2), TR_DIVINE_SPARK (GOD:3) — where lacking the treasure inflicts a crushing **-30% breakthrough penalty (-0.30 in BN.mod)**, while holding it grants **+16% (+0.16 swing = 46% total delta!)**.
4. **Runtime Preparation & Allocation Engines**:
   - BN.daily, BN.decide, BN.prepare, BN.pillMult, BN.onResult, and ECO.usePill wrapper (js/bottleneck.js), A.breakthrough (js/ai.js), SECT.AUTOMATION.autoDistributeBreakthroughPills (js/sect.js), and CULT.breakthroughChance (js/cultivation.js).

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2. Critical Defect 1: [PILL-BT-01] 31 Grand Dao Paths (72% of All Paths) Have ZERO P:BT_<PATH>_<STAGE> Recipes & ZERO BT.TREASURES — Generating Phantom btDemand & Starving Rank 0–1 Disciples!

2.1 Exact Code Proof (js/bottleneck.js:13–14, 79–106, 121–127 & data/data_bt.js)

1. BN.pillKey (bottleneck.js:14) hardcodes:

```javascript
BN.pillKey = c => 'P:BT_' + c.path + '_' + c.stage;
```

2. Every day in BN.daily(W) (bottleneck.js:79), for **every** bottlenecked disciple on CHRONO_EPOCH, SHADOW_SAMURAI, ASTRO_DIVINATION, BUDDHA_DHARMA, VOID_SPACE, MERCANTILE_DAO, DIVINE_SUMMONER, OCCULT_MAGE, NETHER_GRAVE, etc. (**all 31 Grand Dao Paths**, and verify any missing stages in the 7 v2.0 specialist paths):
   - S.btDemand['CHRONO_EPOCH:3'] is populated!
   - ECO.pickOrder in js/economy.js reads W.sect.btDemand, looks up ECO.recipeById['BT_CHRONO_EPOCH_3'] $\rightarrow$ **undefined!** The Alchemy Hall **never brews anything** to satisfy their bottleneck!

- `CPE.pickPurchase` (`js/cpeco.js:278`) checks `if (e.btFor) { if (e.btFor.path === c.path && e.btFor.stage === c.stage ...) }` $\rightarrow$ **never matches** for any of the 31 Grand Dao paths, so disciples on those 31 paths **never buy a breakthrough pill with CP from the Sect Shop**!

3. Now trace what happens when they finally give up waiting (`waited >= limit`) and enter `BN.prepare(W, n)` (`bottleneck.js:121–127`):
   - `BN.heldPill(n)` is **`null`** (no `P:BT_<PATH>_<STAGE>` exists!).
   - `ECO.count(W, BN.pillKey(c))` is **`0`**.
   - Fallback general pill search (`line 124`):

```JavaScript
else if (n.rank >= 2) { // โอสถทั่วไป (ไม่เฉพาะขั้น) จากคลัง
  for (const key of Object.keys(W.sect.stock)) {
    const info = ECO.isPill(key) ? ECO.pillInfo(key) : null;
    if (info && info.effect.breakthroughBonus && !info.effect.btFor && info.tier
<= ... && W.sect.stock[key] > 0) {
      ECO.usePill(W, n, key); break;
    }
  }
}
```

   - **Outer/Regular Disciples (`rank === 0` or `rank === 1`, typically Stages 1 $\to$ 3) are HARD-BLOCKED (`n.rank >= 2` is `false`!)** from taking general breakthrough pills from stock AND couldn't buy one from `CPE.pickPurchase` (`cpeco.js:278` only scores `e.btFor` pills, completely ignoring general `!e.btFor && e.breakthroughBonus` pills in the CP shop!!)!
   - Even for `rank >= 2` Elders, `Object.keys(W.sect.stock)` grabs the **first insertion-order pill** (`break;`) — meaning a Stage-7 Grand Elder attempting Immortal Ascension grabs a **Tier-1/+5% pill** instead of the **Tier-6/+28% Golden Core pill** sitting right behind it in `W.sect.stock`!

4. **Asymmetric Bottleneck Treasure Bias (`data_bt.js:35–43` & `bottleneck.js:25`)**:
   - `XIAN`, `BODY`, and `GOD` suffer **`-30%` breakthrough chance** at major realms unless they find/buy their `BT.TREASURES` (`46%` swing), whereas the other **40 Paths** have `BN.treasureOf(c) === null` (`0%` penalty, `0` treasure quest, `0` wandering-merchant treasure drama)!

2.2 Architectural Fix

1. **Universal Lineage-Mapped Breakthrough Pill & Treasure Resolver (`BN.resolveBreakthroughAid(c)`)**:

- Either procedurally register canonical `BT_<PATH>_<STAGE>` recipes for all remaining paths using existing `MAT_V230_001..024` / regional herbs, **OR** map every path without a bespoke `BT_<PATH>_*` recipe to a **Lineage Pill Family** (e.g., Martial/Physical family `BODY`/`SHADOW_SAMURAI`/`ARENA_GLORY`/`DEVOURING_FLESH`; Dharma/Divine family `GOD`/`BUDDHA_DHARMA`/`DIVINE_*`/`ANCESTRAL_RITUAL`; Astral/Spacetime family `ASTRO_DIVINATION`/`VOID_SPACE`/`CHRONO_EPOCH`/`GRAVITY_DAO`/`PRIMORDIAL_CHAOS`; Soul/Literati/Occult family `HEART_SOUL`/`INK_LITERATI`/`LORE_ARCHIVIST`/`SONIC_HARMONY`/`SOUL_PUPPETRY`/`OCCULT_MAGE`; Nether/Venom family `DEMON`/`VENOM_GU`/`VOID_HERETIC_PACT`/`NETHER_GRAVE`) mapped via `Math.round(CULT.normalizedStage(n))`!
- Likewise, map major realm gates (`equivalentStage` `3–4`, `5–6`, `7–8`) of all 43 paths to thematic `BT.TREASURES` (or a lineage treasure group) so every path shares the authentic Xianxia major-realm treasure hunt!

2. **Fix `CPE.pickPurchase` & `BN.prepare` Fallback**:
   - Allow `CPE.pickPurchase` (`cpeco.js:278`) to buy **general `breakthroughBonus` pills (`!e.btFor`)** when stuck at a bottleneck (`c.bn`) and no exact `btFor` pill is held!
   - Sort candidate fallback pills in `BN.prepare` (`bottleneck.js:124`) by **`info.effect.breakthroughBonus` DESC** (picking the strongest eligible pill for the cultivator's `effectiveTechStage`, not insertion order `[0]`!), and allow `n.inv.pills` general breakthrough pills for **all ranks (`rank >= 0`)**.

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3. Critical Defect 2: `[PILL-BT-02]` Personal-Bag Breakthrough Pills in `BN.prepare` Completely Bypass Drug Resistance (`btTaken`), Pill Quality (`pillQuality`), and Wrong-Path/Stage Multipliers!

3.1 Exact Code Proof (`js/bottleneck.js:122` vs `167–178` & `js/economy.js:346`)

Look at `BN.prepare` (`bottleneck.js:121–123`) vs `ECO.usePill` (`bottleneck.js:167–178`):

- In `BN.daily` (`line 101`) and `CPE.buy` (`cpeco.js:325`), breakthrough pills are placed into **`n.inv.pills[k]`** (the disciple's personal bag).

- When the disciple starts `A.breakthrough.start` $\rightarrow$ `BN.prepare(W, n)` (`line 122`):

```
JavaScript
const c = n.cult, k = BN.heldPill(n);
if (k) { n.inv.pills[k]--; if (n.inv.pills[k] <= 0) delete n.inv.pills[k];
ECO.applyPill(W, n, k); }
```

- Compare what `ECO.applyPill(W, n, k)` (`economy.js:346`) does vs `ECO.usePill(W, n, k)` (`bottleneck.js:167–178`):
    1. **`ECO.applyPill` NEVER calls `BN.pillMult(c, e.btFor)` and NEVER increments `c.btTaken`!**
        - Suppose a disciple fails a breakthrough (`BN.onResult(W, n, false)` at `line 30`: notice `c.btTaken` is **intentionally NOT reset on failure** so repeated pills at the same stage suffer `-30%` diminishing returns `Math.pow(0.7, btTaken)`: `100% -> 70% -> 49% -> 34.3%`!).
        - Because `BN.daily` and `CPE.buy` put pills into `n.inv.pills[k]`, **95%+ of breakthrough pills in real gameplay go through `if (k) { ... ECO.applyPill }` — so `c.btTaken` STAYS `0` FOREVER AND DRUG RESISTANCE (`BN.pillMult`) NEVER TRIGGERS IN LIVE PLAY!!**
    2. **`ECO.applyPill` Ignores Personal Pill Quality (`n.inv.pillQuality[k]`)!**
        - A Supreme-Grade `q = 1.65` Masterpiece Foundation Pill brewed by a Grandmaster Alchemist gives the **exact same** `0.18` (`+18%`) breakthrough bonus as a crude `q = 0.72` novice pill!
    3. **Personal Wrong-Stage / Wrong-Path `P:BT_*` Pills (`e.g. looted from enemies, bought from visiting merchants in external_market.js, or kept after retraining`) Are Never Consumed AND If Consumed via `applyPill` Give `100%` Bonus Instead of `20%` + `×1.8` Toxicity!**
        - Look at `economy.js:346`: `if (e.breakthroughBonus) c.btBonus = Math.max(c.btBonus || 0, e.breakthroughBonus);` — it doesn't even check `e.btFor`! If an NPC or `autoUse` calls `applyPill` on `P:BT_XIAN_8` while at `BODY` Stage 1, they get the **full `+30%` Stage-8 Immortal Ascension bonus** with `1.0x` toxicity instead of `0.2 * 0.6 = 12%` effectiveness and `1.8x` Meridian-Scorching Toxicity!
    4. **Vow Violation in `BN.prepare` (`bottleneck.js:121`):**
        - Remember in `ai.js`, `A.breakthrough.start` was patched by `bottleneck.js:146`: `AB.start = (n, W, ctx) => { BN.prepare(W, n); n.cult.lastTry = W.day; n.sealedBt = true; return true; };` **`bottleneck.js:146` COMPLETELY**

OVERWROTE `ai.js`'s `A.breakthrough.start` (which had the `VOW_NO_MEDICINE` check!) WITH A NEW FUNCTION THAT CALLS `BN.prepare(W, n)` UNCONDITIONALLY WITHOUT CHECKING `VOW_NO_MEDICINE` OR `VOW_ABSTAIN_PILLS`!! Let that sink in: whoever added `VOW_NO_MEDICINE` to `ai.js`'s `A.breakthrough.start` didn't realize `bottleneck.js` (loaded *after* `ai.js` in `load.js`!) **overwrites `AI.ACTS.breakthrough.start` completely on line 146**! So every `VOW_NO_MEDICINE` / `VOW_ABSTAIN_PILLS` cultivator **still gets force-fed a breakthrough pill in `BN.prepare` right before their breakthrough, shattering their vow (`vowBacklash` + `+35 heartDemon`) at the exact moment they face Heavenly Tribulation**!!

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4. Critical Defect 3: `[PILL-BT-03]` The Triple Breakthrough-Bonus Variable Split (`c.btBonus` vs `npc._btBonusPct` vs Buff `b.btBonus`) & `estimate()` Preview Desync!

4.1 Exact Code Proof

Across the codebase, **four different fields** store breakthrough probability bonuses:

1. **`npc.cult.btBonus`** (`economy.js:346`, `bottleneck.js:171`): Written by `P:BT_*` and `REC140_*` pills; cleared on breakthrough completion (`c.btBonus = 0`).
2. **`npc._btBonusPct`** (`economy.js:882` for `EFX_BREAKTHROUGH_BOOST` items/pills, `cultivation.js` for `SUPPORT_ARTS` & `FORBIDDEN_ARTS`):
    - Note: `_btBonusPct` IS in `PERSISTED_UNDERSCORE_KEYS` (`sim.js:385`), **HOWEVER**:
    - Does `BN.decide` / `estimate(W, n)` (`bottleneck.js:35`) include `_btBonusPct` and active buff `b.btBonus` (`clarityLotus +5%`, `meridianUnblock +8%`, `mentorGuidance +8%`, `alchemicalHarmony +6%` in `stats.js:95–119`)?
    - **Check `stats.js:95–119` (`BUFF_CATALOG`):** `clarityLotus`, `meridianUnblock`, `mentorGuidance`, and `alchemicalHarmony` all declare `btBonus: 0.05..0.08`, and `ST.addBuff` (`stats.js:235`) copies `btBonus: b.btBonus || 0` onto `n.buffs` — **verify in `CULT.breakthroughChance` (`cultivation.js`) that `n.buffs` (`b.until > W.t && b.btBonus`), `npc._btBonusPct`, AND `c.btBonus` are ALL summed cleanly AND decayed/consumed on `BN.onResult` so**

`_btBonusPct` **from a one-time** `EFX_BREAKTHROUGH_BOOST` **pill does NOT persist permanently across all future stages 1 $\to$ 9!**

- **WAIT — LOOK AT** `economy.js:882` **(**`EFX_BREAKTHROUGH_BOOST`**):** `npc._btBonusPct = Math.min(0.35, (npc._btBonusPct || 0) + (p.bonusChance || 0.12));` And look at `BN.onResult` (`bottleneck.js:28–30`): `if (ok) { c.found = 0; c.failExp = 0; c.btTaken = 0; c.treasure = null; c.bn = null; c.failN = 0; }` **`BN.onResult` NEVER RESETS `npc._btBonusPct`!!** Once a disciple swallows **3 Tier-2** `EFX_BREAKTHROUGH_BOOST` **pills** at Stage 2, **`npc._btBonusPct = 0.35` (`+35%` breakthrough chance!) STAYS PERMANENTLY LOCKED ON THAT NPC FOR STAGES 3, 4, 5, 6, 7, 8, AND 9 FOR THE REST OF THEIR IMMORTAL LIFE**!!

3. `estimate(W, n)` **Double-Counts or Misestimates Pill Bonus (**`bottleneck.js:35`**):**
   - `estimate` computes `p0 = CULT.breakthroughChance(n, { env: 0.05, pill: c.btBonus || 0 }) + heldBonus` — assuming `env: 0.05` (a basic mat!) even when the Sect has a **Lv.5 Meditation Hall + Peak Sanctuary Array + Closed-Door Seclusion Chamber (`env` up to `0.22+`)**, AND doesn't multiply `heldBonus` by `BN.pillMult(c, heldInfo.effect.btFor)` (so if `btTaken === 2`, `estimate` thinks the held pill gives `+18%` when it actually gives `+8.8%`, tricking cautious NPCs into attempting fatal breakthroughs prematurely!).

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5. Logic & Autonomous World Flaws: `[PILL-BT-04]` Desperate Wrong-Stage Pill Gambling, Hoarded Unused `P:BT_*` Pills After Advancement, & `autoDistributeBreakthroughPills`

1. **Emergent Desperation Mechanic Exists in `BN.pillMult` (`bottleneck.js:18–22`) — Yet Zero NPCs Ever Use It!**:
   - Notice the brilliant design comment at the top of `bottleneck.js:17`: `/* ตรงขั้น/สาย = เต็ม; ขั้นต่ำกว่า 30%; ขั้นสูงกว่า 60% (พิษ ×1.8); ผิดสาย 20%; ดื้อยาลง 30% ต่อเม็ด */`
   - Someone carefully coded `BN.pillMult` for **desperate cultivators taking an over-tier or off-path breakthrough pill at `×1.8` toxicity risk**, **yet `BN.heldPill` (`line 15`) and `BN.prepare` (`line 121`) ONLY ever check exact `P:BT_<PATH>_<STAGE>`**!
   - **Emergent Opportunity**: When an NPC has **`urgent === true` (lifespan `< 12%` remaining — about to die of old age!)** or **`ambition > 80 && riskTolerance > 75`**, and no exact-stage pill exists, let them **autonomously gamble on a higher-stage or off-path `P:BT_*` pill in their bag/shop**, suffering

the `×1.8` toxicity & meridian strain in exchange for one last shot at longevity! Conversely, disciplined/cautious disciples refuse mismatched pills.

2. **Stale `P:BT_<PATH>_<OLD_STAGE>` Pills Rotting in Disciples' Pockets Forever**:
   - If a disciple buys `P:BT_XIAN_2` (or receives two copies), breaks through to Stage 3 without needing the second copy, `P:BT_XIAN_2` sits in `n.inv.pills` forever because `pillNeed` (`external_market.js:378`) returns `0`, `A.donate` (`cpeco.js:211`) **never donates personal `n.inv.pills` (`only mats, techs, recipes, gear`!)**, and `CPE.autoUse` never touches `btFor` pills!
   - **Fix**: Allow `A.donate` (or a new junior-disciple gift hook in `COG`) to **donate obsolete lower-stage `P:BT_*` pills back to the Sect Treasury for CP — OR gift them directly to a bottlenecked junior apprentice (`n.apprenticeIds` / `disciples`) on the same path**! Imagine a newly ascended Elder handing their leftover Foundation Pill to their struggling junior sibling or personal disciple—pure emergent Wuxia storytelling!

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6. Clean Remediation Blueprint for GPT & Gemini

1. **Centralize Breakthrough Pill Application (`BN.consumeBreakthroughPill(W, npc, key, quality, source)`)**:
   - Route **both** personal-bag pills (`n.inv.pills[k]`) and sect-stock pills through one function that:
     1. Checks `CULT.isPillForbiddenByVow(npc)` FIRST — if forbidden and `!npc._desperateBreakVow`, skip pill consumption cleanly instead of shattering the cultivator's vow right before Tribulation!
     2. Scales `rawBonus = info.effect.breakthroughBonus * (0.75 + 0.25 * clamp(quality || 1, 0.7, 1.75))`.
     3. Applies `mult = info.effect.btFor ? BN.pillMult(npc.cult, info.effect.btFor) : Math.pow(0.75, npc.cult.btTaken || 0)`.
     4. Updates `npc.cult.btBonus = Math.min(0.48, Math.max(npc.cult.btBonus || 0, rawBonus * mult))` (or additive diminishing stack capped at `0.48`), increments `npc.cult.btTaken++`, and applies `toxicity * (overStageOrWrongPath ? 1.8 : 1.0)`.
2. **Unify Breakthrough Bonus Fields & Reset Lifecycle**:
   - Separate **permanent talent/trait bonuses** (`npc.ext.permBtBonus`) from **single-attempt preparation bonuses** (`npc.cult.btBonus` + `npc._btBonusPct`), and **reset `npc._btBonusPct = 0` inside `BN.onResult(W, npc, ok)`** (preserving only active aura contributions recomputed by `SUPPORT_ARTS` / `FORBIDDEN_ARTS`) so 3 cheap

`EFX_BREAKTHROUGH_BOOST` pills cannot grant permanent `+35%` breakthrough chance for life!

3. **Cover All 43 Paths in `BN.pillKey` & `BT.TREASURES`**:
   - Provide Lineage Breakthrough Pills (`BT_LINEAGE_*_1..8`) or path-mapped aliases so all 43 paths have craftable, purchasable, and distributable bottleneck pills and major-realm treasures.
4. **Junior Gifting & Obsolete `P:BT_*` Circulation**:
   - When `c.stage > btFor.stage`, allow loyal/empathetic seniors to gift obsolete `P:BT_*` pills to bottlenecked disciples/juniors (`+affection`, `+gratitude`, biography entry), or donate them to `W.sect.stock` for CP.

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7. Mandatory Verification Suite Additions

1. **Personal-Bag Resistance & Quality Test**: Give an NPC three `P:BT_XIAN_2` pills in `n.inv.pills` across 3 failed attempts; assert `c.btTaken` increments `0 -> 1 -> 2 -> 3` and effective `btBonus` scales by `1.0x -> 0.7x -> 0.49x` and reflects `pillQuality`.
2. **Vow Protection in `A.breakthrough.start` (`bottleneck.js:146`)**: Give an NPC with `VOW_NO_MEDICINE` or `VOW_ABSTAIN_PILLS` a `P:BT_XIAN_2` in `n.inv.pills` and start `A.breakthrough`; assert the pill is **not** consumed and the vow remains unbroken.
3. **`_btBonusPct` Single-Attempt Decay Test**: Consume `EFX_BREAKTHROUGH_BOOST`, complete 1 breakthrough (`BN.onResult`), and assert `_btBonusPct` resets to `0` for the next realm.
4. **43-Path Bottleneck Coverage Test**: For all 43 paths in `CULT.PATHS`, assert `BN.resolveBreakthroughPillKey(c)` resolves to an existing craftable recipe in `ECO.recipeById`!